

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

If $\tan A = \frac{1}{\sqrt{x(x^2+x+1)}}$, $\tan B = \frac{\sqrt{x}}{\sqrt{x^2+x+1}}$ and $\tan C = (x^{-3} + x^{-2} + x^{-1})^{\frac{1}{2}}$, $0 < A, B, C < \frac{\pi}{2}$, then $A + B$

is equal to :

- (1) C
- (2) $\pi - C$
- (3) $2\pi - C$
- (4) $\frac{\pi}{2} - C$

Q2 - 2024 (30 Jan Shift 2)

For $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, let $3 \sin(\alpha + \beta) = 2 \sin(\alpha - \beta)$ and a real number k be such that $\tan \alpha = k \tan \beta$. Then the value of k is equal to :

- (1) $-\frac{2}{3}$
- (2) -5
- (3) $\frac{2}{3}$
- (4) 5

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Questions with Answer Keys

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Solutions

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Q1

Finding $\tan(A + B)$ we get

$$\Rightarrow \tan(A + B) =$$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{\frac{1}{\sqrt{x(x^2+x+1)}} + \frac{\sqrt{x}}{\sqrt{x^2+x+1}}}{1 - \frac{1}{x^2+x+1}}$$

$$\Rightarrow \tan(A + B) = \frac{(1+x) \left(\sqrt{x^2+x+1} \right)}{(x^2+x)(\sqrt{x})}$$

$$\frac{(1+x) \left(\sqrt{x^2+x+1} \right)}{(x^2+x)(\sqrt{x})}$$

$$\tan(A + B) = \frac{\sqrt{x^2+x+1}}{x\sqrt{x}} = \tan C$$

$$A + B = C$$

Q2

$$3 \sin \alpha \cos \beta + 3 \sin \beta \cos \alpha$$

$$= 2 \sin \alpha \cos \beta - 2 \sin \beta \cos \alpha$$

$$5 \sin \beta \cos \alpha = -\sin \alpha \cos \beta$$

$$\tan \beta = -\frac{1}{5} \tan \alpha$$

$$\tan \alpha = -5 \tan \beta$$

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